Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/16/2025
PI Name:	Tahimic, Candice Ginn Ph.D.		
Project Title:	Cardiovascular Responses to Simulated Spaceflight: Molecular Signatures and Surrogate Outputs to Measure CVD Risk		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (2) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32224	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B
Start Date:	01/28/2021	**End Date:**	12/31/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	7
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2023 per NSSC information (Ed., 1/6/24). NOTE: End date changed to 03/31/2023 per NSSC information (Ed., 5/18/23).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Goukassian, David M.D., Ph.D. (ICAHN School of Medicine at Mount Sinai) Ronca, April Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	80NSSC21K0548		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In this ground-based rodent study, we aim to systematically define molecular signatures of cardiovascular performance across doses of acute simulated galactic cosmic radiation (Five-ion GCR) at early and late timepoints post-exposure. We also will determine the contribution of biological sex and the combined effects of GCR and microgravity on clinically relevant and emerging measures of cardiovascular health. We hypothesize that exposure to space radiation alone or in combination with microgravity results in early and late changes in the structure, transcriptome, redox signaling, and cytokine milieu of cardiovascular tissue, some of which have known links to decreased performance, aging, and increased cardiovascular disease (CVD) risk. We further posit that other less invasive clinically relevant measures of immune, behavior, and neuromotor function will be informative towards extrapolating the effects of deep space missions on human cardiovascular health. To achieve the project goals and test the hypothesis, we will take advantage of a rare tissue sharing opportunity from a recently funded Human Research Program (HRP) study. The experiment design of this funded investigation includes a GCR dosing study on crew age-matched female and male mice (6 months old) as well as combined exposure study with simulated microgravity. A comprehensive panel of outcomes will be assessed in the funded study and includes measures of immune health, brain molecular and structural changes, behavior, anxiety, cognition, and neuromotor function. Our proposed approach is to measure clinically relevant indicators of cardiovascular performance and perform transcriptomic profiling by RNAseq to determine dose and time-dependency of cardiovascular responses. To further facilitate extrapolation of results to humans, rodent RNAseq data will be compared to publicly available human RNAseq datasets from aging and CVD progression studies. The results from analysis of rodent cardiovascular tissue also will be compared to corresponding blood data to link immune and cardiovascular changes. Further, cardiovascular findings will be interpreted in light of behavioral testing results to gain insight on any relationships between cardiovascular outcomes and changes in neuromotor, anxiety levels, and cognitive performance. The significance/impact of this study is that it contributes to increased understanding on the mechanisms of degenerative changes in cardiovascular tissue and the clinical endpoints they suggest. This study also is expected to provide insight on the latency period for radiation-induced cardiovascular changes and any sex differences in these outcomes. Lastly, our findings are expected to generate testable hypotheses for the development of countermeasures and less invasive surrogate biomarkers to monitor cardiovascular health in-flight and after return.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study is expected to contribute to increased understanding on the mechanisms of spaceflight-induced degenerative changes in cardiovascular tissue and the clinical endpoints they suggest. This study also is expected to provide insight on the latency period for radiation-induced cardiovascular changes and any sex differences in these outcomes. Lastly, our findings are expected to generate testable hypotheses for the development of countermeasures and less invasive surrogate biomarkers to monitor cardiovascular health in-flight and after return. Cardiovascular deconditioning observed in spaceflight resemble aspects of cardiovascular aging and disease on Earth. Hence, this study also may prove informative in the development new therapies for cardiovascular disease on Earth.
Task Progress:	Spaceflight leads to cardiovascular deconditioning in the absence of mitigation strategies. Some of the cardiovascular changes in response to spaceflight are attributed to altered microgravity levels which cause a net movement of fluids towards the head and thoracic region. During extended space missions beyond low Earth orbit, astronauts will encounter prolonged periods of weightlessness and low dose space radiation. The main objective of this study is to define the molecular signatures of the cardiovascular response to simulated space radiation, or in combination with simulated microgravity, across timepoints relevant to mission duration and recovery. In addition, we aim to determine whether sex impacts cardiovascular responses to these spaceflight factors. Our overarching goals of this study are to improve understanding of the cardiovascular risks associated with long duration spaceflight and to inform on the development of countermeasures to maintain crew health during and after spaceflight. To meet these objectives, we make use of a rodent model for simulated spaceflight. We conducted a series of experiments to gain insight on how the heart and vascular tissue respond to exposure to simulated space radiation and simulated microgravity alone or in combination. We found that simulated space radiation can lead to long-term changes in the expression of genes that play a role in the development of cardiovascular disease. In addition, simulated space radiation can have long lasting effects on the length of telomeres, one of the indicators of cellular aging. We also find that the heart of females and males show differences in their responses to simulated space radiation. Further, simulated space radiation alone or in combination with simulated microgravity can alter the interactions between genes that function in the inflammatory process. Taken together, our findings reveal that simulated space radiation can lead to long-term changes in the expression of genes implicated in the development of cardiovascular disease which should be considered in the development of post-flight cardiovascular reconditioning strategies and interventions to promote healthy aging of the cardiovascular system. Lastly, our findings highlight the need for sex-specific strategies in monitoring and maintaining cardiovascular health during and after deep space missions.
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